

1. MC.04.01.ALGO (Algorithmic)

Which of the following statements is true?

- ✓
- ☐

a. Job-order costing is ideal for homogeneous products that are produced in large volumes.
- ☒

b. The job cost sheet is subsidiary to the work-in-process account.
- ☐

c. Process costing is used only for services.
- ☐

d. A work-in-process inventory account is not used when using process costing.
- ☐

e. All of these.

Feedback

Check My Work

1) Job-order costing is used for products and services that differ from each other. Jobs vary by cost and work performed. Costs are accumulated by job. Work in Process accounts for each job.

2) Process costing is used by mass-produced products or services. Production costs are accumulated by process or department for a period of time. Work in Process accounts for each process and/or department.

Solution
b

2. MC.04.02.ALGO (Algorithmic)

The ending balance of which of the following accounts is calculated by summing the totals of the open (unfinished) job-order cost sheets?

- ✓
- ☐

a. Unallocated Manufacturing Overhead
- ☐

b. Cost of Goods Sold
- ☐

c. Total Fixed Costs
- ☐

d. Accounts Payable
- ☒

e. Work in Process

Feedback

Check My Work

Materials are inventory to be used in production. Work in process consists of all incomplete jobs. Finished goods consist of completed but unsold jobs. Overhead is applied to work in process as determined. Cost of goods sold is finished goods inventory that is sold.

Solution
e

3. MC.04.03

In a normal costing system, the cost of a job includes

- ✓
- ☒

a. actual direct materials, actual direct labor, and estimated (applied) overhead.
- ☐

b. estimated direct materials, estimated direct labor, and estimated overhead.
- ☐

c. actual direct materials, actual direct labor, actual overhead, and actual selling cost.

- ☐ d. actual direct materials, actual direct labor, and actual overhead.
- ☐ e. none of these. Job-order costing requires the use of actual, not normal, costing.

Feedback

Check My Work

Actual costing consists of actual costs for direct materials, direct labor, and overhead. Normal costing consists of actual costs for direct materials; direct labor and applied overhead costs are based on estimated amounts.

Solution
a

4. MC.04.04.ALGO (Algorithmic)

The predetermined overhead rate equals

- ✓ ☒ a. estimated overhead divided by estimated activity level for a period.
- ☐ b. actual overhead multiplied by actual activity level for a period.
- ☐ c. actual overhead divided by actual direct labor hours.
- ☐ d. actual overhead minus estimated overhead.
- ☐ e. actual overhead divided by estimated overhead for a given period.

Feedback

Check My Work

Predetermined overhead rate is calculated by dividing the estimated overhead costs by the estimated activity level for a period.

Solution
a

5. MC.04.06

Applied overhead is

- ✓ ☒ a. an important part of normal costing.
- ☐ b. never used in normal costing.
- ☐ c. an important part of actual costing.
- ☐ d. the predetermined overhead rate multiplied by estimated activity level.
- ☐ e. none of these.

Feedback

Check My Work

Applied overhead is the amount of overhead determined using the predetermined overhead rate used in product/job cost calculations.

Solution
a

6. MC.04.07.ALGO (Algorithmic)

The overhead variance is overapplied if



- ☐ a. the predetermined overhead rate is greater than the actual overhead rate.
- ☐ b. estimated overhead is less than applied overhead.
- ☒ c. actual overhead is less than applied overhead.
- ☐ d. estimated overhead is more than applied overhead.
- ☐ e. applied overhead is more than total manufacturing costs.

Feedback

Check My Work

Compare applied overhead to actual overhead costs incurred.

Solution
c

7. MC.04.09

Which of the following is typically a job-order costing firm?



- ☐ a. Paint manufacturer
- ☐ b. Pharmaceutical manufacturer
- ☐ c. Cleaning products manufacturer
- ☐ d. Cement manufacturer
- ☒ e. Large regional medical center

Feedback

Check My Work

Job-order costing is used for products and services that differ from each other. Jobs vary by cost and work performed.

Solution
e

8. MC.04.10

Which of the following is typically a process-costing firm?



- ☒ a. Paint manufacturer
- ☐ b. Custom cabinetmaker
- ☐ c. Large regional medical center
- ☐ d. Law office
- ☐ e. Custom framing shop

Feedback

Check My Work

Process costing is used by mass-produced products or services. Production costs are accumulated by process or department for a period of time.

Solution

a

9. MC.04.13

The costs of a job are accounted for on the

- ☐ a. materials requisition sheet.
- ☐ b. time ticket.
- ☐ c. requisition for overhead application.
- ☐ d. sales invoice.
- ☒ e. job-order cost sheet.



Feedback

Check My Work

A subsidiary to the work-in-process accounts; the primary document to accumulate all costs per job.

Solution

e

10. MC.04.14

Wilson Company has a predetermined overhead rate of \$5 per direct labor hour. The job-order cost sheet for Job 145 shows 500 direct labor hours costing \$10,000 and materials requisitions totaling \$17,500. Job 145 had 1,000 units completed and transferred to Finished Goods. What is the cost per unit for Job 145?

- ☐ a. \$20
- ☐ b. \$17.50
- ☐ c. \$25
- ☒ d. \$30
- ☐ e. \$22,500



Feedback

Check My Work

Multiply overhead rate by the number of direct labor hours (driver or activity). Add overhead cost to direct materials and direct labor cost for total job cost. Divide total job cost by number of units completed for per unit cost.

Post-Submission

Direct materials	\$17,500
Direct labor	10,000
Applied overhead (\$5 × 500 DLH)	2,500
Total job cost	<u>\$30,000</u>
÷ Number of units	<u>1,000</u>
Unit cost	<u>\$ 30</u>

Solution
d

11. MC.04.18

(Appendix 4B) An example of a producing department is



- ☐ a. a materials storeroom.
- ☐ b. the maintenance department.
- ☐ c. engineering design.
- ☒ d. assembly.
- ☐ e. All of these.

Feedback

Check My Work

Producing departments are directly responsible for creating products and/or services to be sold.

Solution
d

12. MC.04.19

(Appendix 4B) An example of a support department is



- ☐ a. data processing.
- ☐ b. personnel.
- ☐ c. a materials storeroom.
- ☐ d. payroll.
- ☒ e. All of these.

Feedback

Check My Work

Support departments are necessary for the overall business but are not directly involved with the production of the products and/or services.

Solution
e

13. EX.04.37

Job-Order Costing versus Process Costing

Required:

Identify each of the following types of businesses as either job-order or process costing.

- a. Hospital services Job-order
- b. Custom cabinet making Job-order

c. Toy manufacturing	<u>Process</u> ▼	✓
d. Soft-drink bottling	<u>Process</u> ▼	✓
e. Airplane manufacturing (e.g., 767s)	<u>Job-order</u> ▼	✓
f. Personal computer assembly	<u>Process</u> ▼	✓
g. Furniture making (e.g., computer desks sold at discount stores)	<u>Process</u> ▼	✓
h. Custom furniture making	<u>Job-order</u> ▼	✓
i. Dental services	<u>Job-order</u> ▼	✓
j. Paper manufacturing	<u>Process</u> ▼	✓
k. Nut and bolt manufacturing	<u>Process</u> ▼	✓
l. Auto repair	<u>Job-order</u> ▼	✓
m. Architectural services	<u>Job-order</u> ▼	✓
n. Landscape design services	<u>Job-order</u> ▼	✓
o. Flashlight manufacturing	<u>Process</u> ▼	✓

Feedback

Check My Work

1) Job-order costing is used for products and services that differ from each other. Jobs vary by cost and work performed. Costs are accumulated by job. Work in Process accounts for each job.

2) Process costing is used by mass-produced products or services. Production costs are accumulated by process or department for a period of time. Work in Process accounts for each process and/or department.

Solution

Job-Order Costing versus Process Costing

Required:

Identify each of the following types of businesses as either job-order or process costing.

a. Hospital services	Job-order
b. Custom cabinet making	Job-order
c. Toy manufacturing	Process
d. Soft-drink bottling	Process
e. Airplane manufacturing (e.g., 767s)	Job-order
f. Personal computer assembly	Process
g. Furniture making (e.g., computer desks sold at discount stores)	Process
h. Custom furniture making	Job-order

- i. Dental services
- j. Paper manufacturing
- k. Nut and bolt manufacturing
- l. Auto repair
- m. Architectural services
- n. Landscape design services
- o. Flashlight manufacturing

Job-order

Process

Process

Job-order

Job-order

Job-order

Process

14. EX.04.39.ALGO (Algorithmic)

Calculating the Predetermined Overhead Rate, Applying Overhead to Production

At the beginning of the year, Debion Company estimated the following:

Overhead \$653,100

Direct labor hours 95,750

Debion uses normal costing and applies overhead on the basis of direct labor hours. For the month of March, direct labor hours were 22,023.

Required:

1. Calculate the predetermined overhead rate for Debion. **Round your answer to the nearest cent.**

\$ 

6.82

per direct labor hour

2. Calculate the overhead applied to production in March. **Round your intermediate calculations to two decimal places and round your answer to nearest whole value.**

\$ 

150,197

Feedback

Check My Work

1. Predetermined overhead rate is an estimate based on dividing the estimated overhead costs by the estimated activity level.

2. Multiply overhead rate by the associated driver.

Post-Submission

1. Predetermined Overhead Rate = $\$653,100 / 95,750 \text{ DLH} = \$6.82 \text{ per direct labor hour}$

2. Applied Overhead = $\$6.82 \times 22,023 \text{ DLH} = \$150,197$

Calculating the Predetermined Overhead Rate, Applying Overhead to Production

At the beginning of the year, Debion Company estimated the following:

Overhead \$653,100

Direct labor hours 95,750

Debion uses normal costing and applies overhead on the basis of direct labor hours. For the month of March, direct labor hours were 22,023.

Required:

1. Calculate the predetermined overhead rate for Debion. **Round your answer to the nearest cent.**

\$ per direct labor hour

2. Calculate the overhead applied to production in March. **Round your intermediate calculations to two decimal places and round your answer to nearest whole value.**

\$

15. EX.04.40.ALGO (Algorithmic)

Calculating the Predetermined Overhead Rate, Applying Overhead to Production, Reconciling Overhead at the End of the Year, Adjusting Cost of Goods Sold for Under- and Overapplied Overhead

At the beginning of the year, Tseng Company estimated the following:

Overhead \$210,000

Direct labor hours 70,000

Tseng uses normal costing and applies overhead on the basis of direct labor hours. For the month of January, direct labor hours were 8,250. By the end of the year, Tseng showed the following actual amounts:

Overhead \$216,000

Direct labor hours 69,600

Assume that unadjusted Cost of Goods Sold for Tseng was \$276,000.

Required:

1. Calculate the predetermined overhead rate for Tseng. **Round your answers to the nearest cent, if rounding is required.**

\$ 

3

per direct labor hour

2. Calculate the overhead applied to production in January. (**Note: Round to the nearest dollar, if rounding is required.**)

\$

24750

3. Calculate the total applied overhead for the year.

\$

208800

Was overhead over- or underapplied? By how much?

Underapplied overhead \$

7200

4. Calculate adjusted Cost of Goods Sold after adjusting for the overhead variance.

\$

283200

Feedback

Check My Work

1. Predetermined overhead rate equals the estimated overhead costs divided by the estimated activity level.
2. Multiply overhead rate by the associated driver.
3. Multiply overhead rate by the amount of driver for the year. Compare applied overhead to actual overhead costs incurred.
4. Adjust Cost of Goods Sold by amount of over(under)applied overhead.

Post-Submission

1. Predetermined Overhead Rate = $\$210,000 / 70,000 \text{ DLH} = 3.00 \text{ per direct labor hour}$
2. Applied Overhead = $\$3.00 \times 8,250 \text{ DLH} = \$24,750$
3. Applied Overhead = $\$3.00 \times 69,600 \text{ DLH} = \$208,800$

Actual overhead	\$216,000
Applied overhead	208,800
Underapplied overhead	<u><u>\$7,200</u></u>

4.

Adjusted Cost of Goods Sold = $\$276,000 + \$7,200 = \$283,200$

Calculating the Predetermined Overhead Rate, Applying Overhead to Production, Reconciling Overhead at the End of the Year, Adjusting Cost of Goods Sold for Under- and Overapplied Overhead

At the beginning of the year, Tseng Company estimated the following:

Overhead \$210,000

Direct labor hours 70,000

Tseng uses normal costing and applies overhead on the basis of direct labor hours. For the month of January, direct labor hours were 8,250. By the end of the year, Tseng showed the following actual amounts:

Overhead \$216,000

Direct labor hours 69,600

Assume that unadjusted Cost of Goods Sold for Tseng was \$276,000.

Required:

1. Calculate the predetermined overhead rate for Tseng. **Round your answers to the nearest cent, if rounding is required.**

\$ per direct labor hour

2. Calculate the overhead applied to production in January. (**Note: Round to the nearest dollar, if rounding is required.**)

\$

3. Calculate the total applied overhead for the year.

\$

Was overhead over- or underapplied? By how much?

overhead \$

4. Calculate adjusted Cost of Goods Sold after adjusting for the overhead variance.

\$

16. EX.04.41.ALGO (Algorithmic)

Calculating Departmental Overhead Rates and Applying Overhead to Production

At the beginning of the year, Glaser Company estimated the following:

	Assembly Department	Testing Department	Total
Overhead	\$715,000	\$729,300	\$1,444,300
Direct labor hours	124,500	64,740	189,240
Machine hours	83,500	65,130	148,630

Glaser uses departmental overhead rates. In the assembly department, overhead is applied on the basis of direct labor hours. In the testing department, overhead is applied on the basis of machine hours. Actual data for the month of March are as follows:

	Assembly Department	Testing Department	Total
Overhead	\$92,950	\$94,809	\$187,759
Direct labor hours	16,185	8,416	24,601
Machine hours	10,855	8,467	19,322

Required:

1. Calculate the predetermined overhead rates for the assembly and testing departments. **Round your answers to the nearest cent.**

	\$	✓
Assembly department overhead rate	5.74	
	per direct labor hour	
	\$	✓
Testing department overhead rate	11.20	
	per machine hour	

2. Calculate the overhead applied to production in each department for the month of March. **If required, round your answers to nearest whole value.**

	\$	✓
Assembly department	92902	
	\$	✓
Testing department	94830	

3. By how much has each department's overhead been overapplied? Enter the amount and then select either "Overapplied" or "Underapplied." **If applicable, enter overapplied amounts beginning with a minus (–) sign.**

	\$	✓		✓
Assembly department	48		<u>Underapplied</u>	
	\$	✓	<u>Overapplied</u>	✓
Testing department				

Feedback

Check My Work

1. Predetermined overhead rate equals the estimated overhead costs divided by the estimated activity level. Find the overhead rate for each department.
2. Multiply overhead rate in each department by the department's driver or activity.
3. Compare applied overhead to actual overhead costs incurred to determine under/over applied overhead.

Post-Submission

1. Assembly Department Overhead Rate = $\$715,000/124,500 \text{ DLH} = \$5.74 \text{ per direct labor hour}$

Testing Department Overhead Rate = $\$729,300/65,130 \text{ mhrs} = \$11.20 \text{ per machine hour}$

2. Assembly Department Applied Overhead = $\$5.74 \times 16,185 = \$92,902$

Testing Department Applied Overhead = $\$11.20 \times 8,467 = \$94,830$

3.

	Assembly Department	Testing Department
Actual overhead	\$92,950	\$94,809
Less: Applied overhead	92,902	94,830
Overhead variance	\$ 48 Underapplied	\$ -21 Overapplied

Solution

Calculating Departmental Overhead Rates and Applying Overhead to Production

At the beginning of the year, Glaser Company estimated the following:

	Assembly Department	Testing Department	Total
Overhead	\$715,000	\$729,300	\$1,444,300
Direct labor hours	124,500	64,740	189,240
Machine hours	83,500	65,130	148,630

Glaser uses departmental overhead rates. In the assembly department, overhead is applied on the basis of direct labor hours. In the testing department, overhead is applied on the basis of machine hours. Actual data for the month of March are as follows:

	Assembly	Testing	
	Department	Department	Total
Overhead	\$92,950	\$94,809	\$187,759
Direct labor hours	16,185	8,416	24,601
Machine hours	10,855	8,467	19,322

Required:

1. Calculate the predetermined overhead rates for the assembly and testing departments. **Round your answers to the nearest cent.**

Assembly department overhead rate \$ per direct labor hour

Testing department overhead rate \$ per machine hour

2. Calculate the overhead applied to production in each department for the month of March. **If required, round your answers to nearest whole value.**

Assembly department \$

Testing department \$

3. By how much has each department's overhead been overapplied? Enter the amount and then select either "Overapplied" or "Underapplied." **If applicable, enter overapplied amounts beginning with a minus (–) sign.**

Assembly department \$

Testing department \$

17. EX.04.42.ALGO (Algorithmic)

Job-Order Costing Variables

On July 1, Job 46 had a beginning balance of \$750. During July, prime costs added to the job totaled \$660. Of that amount, direct materials were four times as much as direct labor. The ending balance of the job was \$1,600.

Required:

1. What was overhead applied to the job during July?

\$ ☒

190

2. What were direct materials for Job 46 for July? Direct labor? **If rounding is required, round your answers to the nearest cent.**

Direct labor	\$	✓
		132.00
Direct materials	\$	✓
		528.00

3. Assuming that overhead is applied on the basis of direct labor cost, what is the overhead rate for the company?

Round your answer to the nearest whole percent.

\$ ✓

144

%

Feedback

Check My Work

1. Find overhead applied by adding beginning balance to prime costs and subtracting from ending balance.
2. Set up equation for prime cost using direct materials and direct labor. Let direct materials equal 4 times direct labor.
3. Divide overhead by the direct labor for the rate.

Post-Submission

1.

Ending Balance = Beginning Balance + Prime Costs + Applied Overhead

\$1,600 = \$750 + \$660 + Applied Overhead

Applied Overhead = \$1,600 - \$750 - \$660 = \$190

2.

Direct Materials = 4 × Direct Labor

Prime Cost = Direct Materials + Direct Labor

\$660 = (4 × Direct Labor) + Direct Labor

Direct Labor = \$660/5 = \$132

Direct Materials = 4 × Direct Labor = 4(\$132) = \$528

3.

Applied Overhead = Direct Labor × Overhead Rate

\$190 = \$132 × Overhead Rate

Overhead Rate = \$190/\$132 = 1.44, or 143.939394%

Job-Order Costing Variables

On July 1, Job 46 had a beginning balance of \$750. During July, prime costs added to the job totaled \$660. Of that amount, direct materials were four times as much as direct labor. The ending balance of the job was \$1,600.

Required:

1. What was overhead applied to the job during July?

\$ 190

2. What were direct materials for Job 46 for July? Direct labor? **If rounding is required, round your answers to the nearest cent.**

Direct labor \$ 132.00

Direct materials \$ 528.00

3. Assuming that overhead is applied on the basis of direct labor cost, what is the overhead rate for the company? **Round your answer to the nearest whole percent.**

\$ 144 %

18. EX.04.44.ALGO (Algorithmic)

Applying Overhead to Jobs, Costing Jobs

Jagjit Company designs and builds retaining walls for individual customers. On August 1, there were two jobs in process: Job 93 with a beginning balance of \$8,650 and Job 94 with a beginning balance of \$7,210. Jagjit applies overhead at the rate of \$6 per direct labor hour. Direct labor wages average \$18 per hour. Data on August costs for all jobs are as follows:

	Job 93	Job 94	Job 95	Job 96
Direct materials	\$910	\$4,490	\$3,320	\$1,250
Direct labor cost	2,160	5,580	2,160	1,080

During August, Jobs 95 and 96 were started. Job 93 was completed on August 17, and the client was billed at cost plus 40%. All other jobs remained in process.

Required:

1. Calculate the number of direct labor hours that were worked on each job in August.

Job 93	✓	120
		DLH
Job 94	✓	

	310
	DLH
	✓
Job 95	120
	DLH
	✓
Job 96	60
	DLH

2. Calculate the overhead applied to each job during the month of August.

Job 93	\$	✓
		720
Job 94	\$	✓
		1,860
Job 95	\$	✓
		720
Job 96	\$	✓
		360

Feedback

Check My Work

1. For each job divide the labor cost by the labor rate.
2. Multiply the applied overhead rate by the labor hours for each job.

Post-Submission

1. The computation for direct labor hours (DHL) for each job is shown below:

- Job 93 = $\$2,160 \div \$18 = 120$ DHL
- Job 94 = $\$5,580 \div \$18 = 310$ DHL
- Job 95 = $\$2,160 \div \$18 = 120$ DHL
- Job 96 = $\$1,080 \div \$18 = 60$ DHL

2. The August applied overhead for each job is shown below:

- Job 93 = $120 \text{ DHL} \times \$6 = \720
- Job 94 = $310 \text{ DHL} \times \$6 = \$1,860$
- Job 95 = $120 \text{ DHL} \times \$6 = \720
- Job 96 = $60 \text{ DHL} \times \$6 = \360

3. Prepare job-order cost sheets for each job as of the end of August. **All amount cells must have an entry; enter zeros as needed.**

Jagjit Company
Job-Order Cost Sheets
August 31

	Job 93	Job 94	Job 95	Job 96
	\$ ✓	\$ ✓	\$ ✓	\$ ✓
Beginning balance	8,650	7,210	0	0
	✓	✓	✓	✓
Direct materials	910	4,490	3,320	1,250
	✓	✓	✓	✓
Direct labor	2,160	5,580	2,160	1,080
	✓	✓	✓	✓
Applied overhead	720	1,860	720	360
	✓	✓	✓	✓
Total	<u>\$ 12,440</u>	<u>\$ 19,140</u>	<u>\$ 6,200</u>	<u>\$ 2,690</u>

Feedback

Check My Work
Correct

4. Calculate the balance in Work in Process on August 31.

The work in process balance on August 31 is \$ ✓

28,030

.

5. What is the price of Job 93?

The price of Job 93 is \$ ✓

17,416

.

6. Conceptual Connection: Partway through the year, Jagjit bought a bulldozer to handle larger jobs. The bulldozer cost \$37,700 and is needed for larger commercial jobs. Smaller residential jobs can still be done with the smaller bobcat tractor.

Complete the statement below that outlines how the bulldozer's cost could be applied to only those jobs that need the larger equipment.

Create a separate department ✓ for the acquisition and use of the bulldozer. Treat the bulldozer's depreciation, fuel, maintenance, etc. as total budgeted cost ✓. Divide this by the anticipated annual hours of use to arrive at the overhead rate ✓.

Feedback

Check My Work

6. Consider the options for applying the cost of the bulldozer to individual jobs.

Post-Submission

4. Work in process, August 31, consists of the following unfinished jobs:

Job 94	\$19,140
Job 95	6,200
Job 96	2,690
Total	<u>\$28,030</u>

5. Price of Job 93 = $\$12,440 + (0.4 \times \$12,440) = \$17,416$

6. Jagjit could treat the acquisition and use of the bulldozer as a separate department and create a departmental overhead rate for it based on the hours used. That is, the overhead rate would be the total budgeted cost of the bulldozer (depreciation, fuel, maintenance, and so on) divided by the anticipated annual hours of use. In this way, only the jobs requiring the use of the heavier equipment would be charged for it.

Feedback

Check My Work

Correct

Solution

1. Calculate the number of direct labor hours that were worked on each job in August.

Job 93		120	DLH
Job 94		310	DLH
Job 95		120	DLH
Job 96		60	DLH

2. Calculate the overhead applied to each job during the month of August.

Job 93	\$	720
Job 94	\$	1,860
Job 95	\$	720
Job 96	\$	360

3. Prepare job-order cost sheets for each job as of the end of August. **All amount cells must have an entry; enter zeros as needed.**

Jagjit Company
Job-Order Cost Sheets

August 31

	Job 93	Job 94	Job 95	Job 96
Beginning balance	\$ 8,650	\$ 7,210	\$ 0	\$ 0
Direct materials	910	4,490	3,320	1,250
Direct labor	2,160	5,580	2,160	1,080
Applied overhead	720	1,860	720	360
Total	\$ 12,440	\$ 19,140	\$ 6,200	\$ 2,690

4. Calculate the balance in Work in Process on August 31.

The work in process balance on August 31 is \$ 28,030 .

5. What is the price of Job 93?

The price of Job 93 is \$ 17,416 .

6. Conceptual Connection: Partway through the year, Jagjit bought a bulldozer to handle larger jobs. The bulldozer cost \$37,700 and is needed for larger commercial jobs. Smaller residential jobs can still be done with the smaller bobcat tractor.

Complete the statement below that outlines how the bulldozer's cost could be applied to only those jobs that need the larger equipment.

Create a separate **department** for the acquisition and use of the bulldozer. Treat the bulldozer's depreciation, fuel, maintenance, etc. as **total budgeted cost** . Divide this by the anticipated annual hours of use to arrive at the **overhead rate** .

19. EX.04.45.ALGO (Algorithmic)

Applying Overhead to Jobs, Costing Jobs

Ochoa Company builds internal conveyor equipment to client specifications. On October 1, Job 877 was in process with a cost of \$18,550 to date.

During October, Jobs 878, 879, and 880 were started. Data on costs added during October for all jobs are as follows:

	Job 877	Job 878	Job 879	Job 880
Direct materials	\$14,480	\$6,040	\$3,440	\$1,780
Direct labor	15,400	8,900	1,900	2,200

Overhead is applied to production at the rate of 65% of direct labor cost. Job 877 was completed on October 28, and the client was billed at cost plus 50%. All other jobs remained in process.

Required:

1. Prepare a brief job-order cost sheet showing the October 1 balances of all four jobs, plus the direct materials and direct labor costs during October. (*Note:* There is no need to calculate applied overhead at this point or to total the costs.)
- All amount cells must have an entry; enter zeros as needed.**

Ochoa Company Job-Order Cost Sheets For the Month of October				
	Job 877	Job 878	Job 879	Job 880
	\$ ✓	\$ ✓	\$ ✓	\$ ✓
Beginning balance	18,550	0	0	0
	✓	✓	✓	✓
Direct materials	14,480	6,040	3,440	1,780
	✓	✓	✓	✓
Direct labor	15,400	8,900	1,900	2,200

Feedback

Check My Work
Correct

2. Calculate the overhead applied to each job during October.

The overhead applied to Job 877 is \$ ✓
10,010
.

The overhead applied to Job 878 is \$ ✓
5,785
.

The overhead applied to Job 879 is \$ ✓
1,235
.

The overhead applied to Job 880 is \$ ✓
1,430
.

3. Calculate the balance in Work in Process on October 31.

The balance in Work in Process for Job 878 is \$ ✓
20,725
.

The balance in Work in Process for Job 879 is \$ ✓

6,575

.

The balance in Work in Process for Job 880 is \$ ✓

5,410

.

The total balance in Work in Process is \$ ✓

32,710

.

4. What is the price of Job 877?

The price of Job 877 is \$ ✓

87,660

.

Feedback

Check My Work

2. Calculate applied overhead and add to cost for each job.
3. Sum the balances in all unfinished jobs.
4. Calculate the cost of the job and add the markup on cost of goods sold.

Post-Submission

2. The applied overhead in October is computed as shown below:

- Job 877 = $\$15,400 \times 0.65 = \$10,010$
- Job 878 = $\$8,900 \times 0.65 = \$5,785$
- Job 879 = $\$1,900 \times 0.65 = \$1,235$
- Job 880 = $\$2,200 \times 0.65 = \$1,430$

3. Work in process, October 31 is determined as shown below:

- Job 878: $\$6,040 + \$8,900 + \$5,785 = \$20,725$
- Job 879: $\$3,440 + \$1,900 + \$1,235 = \$6,575$
- Job 880: $\$1,780 + \$2,200 + \$1,430 = \$5,410$

The total balance in Work in Process is $\$20,725 + \$6,575 + \$5,410 = \$32,710$.

4. The calculations for the cost and price of Job 877 are shown below:

- Cost of Job 877 = $\$18,550 + \$14,480 + \$15,400 + \$10,010 = \$58,440$
- Price of Job 877 = $\$58,440 + (0.5 \times \$58,440) = \$87,660$

Check My Work

Correct

Solution

1. Prepare a brief job-order cost sheet showing the October 1 balances of all four jobs, plus the direct materials and direct labor costs during October. (Note: There is no need to calculate applied overhead at this point or to total the costs.) **All amount cells must have an entry; enter zeros as needed.**

**Ochoa Company
Job-Order Cost Sheets
For the Month of October**

	Job 877	Job 878	Job 879	Job 880
Beginning balance	\$ 18,550	\$ 0	\$ 0	\$ 0
Direct materials	14,480	6,040	3,440	1,780
Direct labor	15,400	8,900	1,900	2,200

2. Calculate the overhead applied to each job during October.

The overhead applied to Job 877 is \$ 10,010 .

The overhead applied to Job 878 is \$ 5,785 .

The overhead applied to Job 879 is \$ 1,235 .

The overhead applied to Job 880 is \$ 1,430 .

3. Calculate the balance in Work in Process on October 31.

The balance in Work in Process for Job 878 is \$ 20,725 .

The balance in Work in Process for Job 879 is \$ 6,575 .

The balance in Work in Process for Job 880 is \$ 5,410 .

The total balance in Work in Process is \$ 32,710 .

4. What is the price of Job 877?

The price of Job 877 is \$ 87,660 .

Derry Company uses job-order costing. At the end of the month, the following information was gathered:

Job # Total Cost Complete? Sold?

301	\$1,520	Yes	No
302	1,220	Yes	Yes
303	730	No	No
304	2,150	Yes	No
305	4,090	Yes	No
306	370	No	No
307	710	Yes	Yes
308	600	No	No
309	1,290	No	No
310	480	No	No

The beginning balance of Finished Goods was \$430, consisting of Job 300, which was not sold by the end of the month.

Required:

1. Calculate the balance in Work in Process at the end of the month.

\$ 
3,470

2. Calculate the balance in Finished Goods at the end of the month.

\$ 
8,190

3. Calculate Cost of Goods Sold for the month.

\$ 
1,930

Feedback

Check My Work

1, 2 & 3. Set up 3 columns and title them "WIP," "Finished Goods," and "Cost of Goods Sold." Enter the cost of each job under the appropriate column. (e.g, an unfinished job belongs under "WIP".) Total the costs in each column.

Post-Submission

1.

Balance in Work in Process (all incomplete jobs):

Job 303	\$ 730
Job 306	370
Job 308	600
Job 309	1,290
Job 310	515
Total	<u>\$3,470</u>

2.

Balance in Finished Goods (all jobs completed but not sold):

Beginning balance (Job 300)	\$ 430
Job 301	1,520
Job 304	2,150
Job 305	4,090
Total	<u>\$8,190</u>

3. Cost of Goods Sold = Job 302 + Job 307

Cost of Goods Sold = \$1,220 + \$710 = \$1,930

Solution

Balance of Work in Process and Finished Goods, Cost of Goods Sold

Derry Company uses job-order costing. At the end of the month, the following information was gathered:

Job # Total Cost Complete? Sold?

301	\$1,520	Yes	No
302	1,220	Yes	Yes
303	730	No	No
304	2,150	Yes	No
305	4,090	Yes	No
306	370	No	No
307	710	Yes	Yes
308	600	No	No
309	1,290	No	No
310	480	No	No

The beginning balance of Finished Goods was \$430, consisting of Job 300, which was not sold by the end of the month.

Required:

1. Calculate the balance in Work in Process at the end of the month.

\$ 3,470

2. Calculate the balance in Finished Goods at the end of the month.

\$ 8,190

3. Calculate Cost of Goods Sold for the month.

\$ 1,930

21. EX.04.47.ALGO (Algorithmic)

Job-Order Cost Sheets, Balance in Work in Process and Finished Goods

Pull Company, a job-order costing firm, worked on three jobs in July. Data are as follows:

	Job 86	Job 87	Job 88
Balance, July 1	\$21,400	\$6,260	\$0
Direct materials	\$10,540	\$12,360	\$16,090
Direct labor	\$15,990	\$12,180	\$23,970
Machine hours	510	290	1,000

Overhead is applied to jobs at the rate of \$15 per machine hour. By July 31, Jobs 86 and 88 were completed. Jobs 82 and 86 were sold. Job 87 remained in process. On July 1, the balance in Finished Goods was \$49,010 (consisting of Job 82 for \$25,630 and Job 84 for \$23,380).

Pull prices its jobs at cost plus 30%. During July, variable marketing expenses were 3% of sales, and fixed marketing expenses were \$2,000; administrative expenses were \$4,770. **(Round all amounts to the nearest dollar.)**

1. Prepare job-order cost sheets for all jobs in process during July, showing all costs through July 31. **All amount cells must have an entry; enter zeros as needed.**

Pull Company Job-Order Cost Sheets For the Month of July

	Job 86	Job 87	Job 88
Balance, July 1	\$ ✓ 21,400	\$ ✓ 6,260	\$ ✓ 0
Direct materials	✓ 10,540	✓ 12,360	✓ 16,090
Direct labor	✓ 15,990	✓ 12,180	✓ 23,970

	✓	✓	✓
Applied overhead	7,650	4,350	15,000
	✓	✓	✓
Total	55,580	35,150	55,060

Feedback

Check My Work

Use the information provided and the overhead rate to fill in the Job Order Cost Sheets.

Review the "How to Prepare Brief Job-Order Cost Sheets" example in the text.

2. Calculate the balance in Work in Process on July 31.

The balance in Work in Process on July 31 is \$ ✓

35,150

.

3. Calculate the balance in Finished Goods on July 31.

The balance in Finished Goods on July 31 is \$ ✓

78,440

.

4. Calculate Cost of Goods Sold for July.

The cost of goods sold for July is \$ ✓

81,210

.

5. Calculate operating income for Prull Company for the month of July.

The operating income for the month of July is \$ ✓

14,426

.

Feedback

Check My Work

Use the Job Order Cost Sheets to determine ending balances in WIP and finished goods based on what jobs are still in process and which jobs are completed and unsold.

Use the Job Order Cost Sheets to determine Cost of Goods Sold.

Use the pricing strategy to determine sales and the information provided for selling and administrative expenses with the preceding amount for cost of goods sold to determine operating income for the period.

Review the "How to Prepare Brief Job-Order Cost Sheets" example in the text.

Post-Submission

2. The Work in Process balance for July 31 equals Job 87 or \$35,150.

3. The balance in Finished Goods for July 31 is determined as follows:

Balance, July 1	Jobs 82 and 84	\$49,010
Jobs transferred in	Jobs 86 and 88	110,640
Jobs sold	Jobs 82 and 86	(81,210)
Ending balance, July 31	Jobs 84 and 88	<u>\$78,440</u>

4. Cost of Goods Sold = Job 82 + Job 86 = \$25,630 + \$55,580 = \$81,210.

5. The operating income for the month of July is computed as shown on the income statement below:

Prull Company
Income Statement
For the Month of July

Sales	\$105,573
Cost of goods sold	<u>81,210</u>
Gross margin	\$24,363
Less:	
Variable marketing expenses	\$3,167
Fixed marketing expenses	2,000
Administrative expenses	<u>4,770</u> 9,937
Operating income	<u>\$14,426</u>

- Sales = \$81,210 + (0.3 × \$81,210) = \$105,573
- Variable market expenses = (0.03 × \$105,573) = \$3,167

Feedback

Check My Work

Correct

Solution

1

1. Prepare job-order cost sheets for all jobs in process during July, showing all costs through July 31. **All amount cells must have an entry; enter zeros as needed.**

Prull Company
Job-Order Cost Sheets
For the Month of July

	Job 86	Job 87	Job 88
Balance, July 1	\$ 21,400	\$ 6,260	\$ 0
Direct materials	10,540	12,360	16,090

Direct labor	15,990	12,180	23,970
Applied overhead	7,650	4,350	15,000
Total	\$ 55,580	\$ 35,150	\$ 55,060

2. Calculate the balance in Work in Process on July 31.

The balance in Work in Process on July 31 is \$ 35,150 .

3. Calculate the balance in Finished Goods on July 31.

The balance in Finished Goods on July 31 is \$ 78,440 .

2

4. Calculate Cost of Goods Sold for July.

The cost of goods sold for July is \$ 81,210 .

5. Calculate operating income for Prull Company for the month of July.

The operating income for the month of July is \$ 14,426 .